



Immunization-Associated Pain: Taking Research to the Bedside

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ABSTRACT

Background: Immunization-associated pain is the number one cause of pain in pediatric settings. Untreated pain has many short- and long-term detrimental effects. The purpose of this project was to implement a nonpharmaceutical immunization-associated—pain prevention program.

Method: This project was implemented at a pediatric primary care clinic. Staff were educated about immunization-associated pain and techniques for decreasing/preventing pain. Families were educated about pain control, and the practitioner and parent/child picked interventions to decrease pain. Preimplementation and postimplementation data were collected using previously validated pain and satisfaction scales.

Results: Pain prevention interventions decreased pain for children between 2 months and 7 years of age by 4.7 points on a Faces, Legs, Activity, Cry, Consolability (i.e., FLACC) scale. For children 7 years and older, pain during immunization decreased on average by 1.76 points on the visual analog scale.

Conclusion: Non-pharmaceutical interventions are effective in decreasing immunization-associated pain. *J Pediatr Health Care.* (2019) 33, 446–454

KEY WORDS

Bundle theory, immunization, nonpharmaceutical, pain

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INTRODUCTION

Immunization-associated pain is the most common type of pain experienced by patients in pediatric health care settings (Wallace, Allen, Lacroix, & Pitner, 2010; Whelan, Kunselman, Thomas, Moore, & Tamburro, 2014). This type of pain often goes unrecognized and is dismissed as short lived (Whelan et al., 2014), but it has numerous short- and long-term effects. These effects include intense reaction to future procedures, higher anxiety levels, increased distress, and higher levels of pain in adult life (MacLaren & Cohen, 2007). The World Health Organization estimates that 12 billion injections are given annually, with approximately 5% of those being childhood vaccines (Taddio et al., 2009).

Background

Pain is defined by the International Association for the Study of Pain as “an unpleasant sensory and emotional experience associated with actual or potential tissue damage” (Taddio et al., 2009). Pain is subjective. Descriptions of pain are based on previous experiences of pain, and early life experiences with pain shape future pain responses (Taddio et al., 2009).

Early pain experiences have long-term sequelae, including central nervous system changes and changes in the responsiveness of the neuroendocrine and immune systems to stress, even into adulthood (Racine et al., 2016). Basic nociceptive connections are present at birth, and the development of pain-processing circuitry in the somatosensory nervous system continues after birth (Taddio et al., 2009). The immaturity of this system and the lack of inhibitory controls for pain may lead to greater responses to painful stimuli in infants and children than in adults (Taddio et al., 2009). Early untreated procedural pain leads to sensitization to future pain because of changes in how the nervous system processes pain (Taddio et al., 2009).

Untreated procedural pain during infancy and early childhood leads to long-term sequelae including anticipatory fear (Racine et al., 2016; Taddio et al., 2009), sensitization to pain

(Hatfield, Meyers, & Messing, 2013), reduced effectiveness of analgesics (Racine et al., 2016), difficulty carrying out procedures, and needle phobia (McMurtry et al., 2015). Anticipatory fear increases both the pain and distress experienced during a procedure, and untreated needle pain leads to needle phobia in approximately 10% of the population (Taddio et al., 2009). Needle phobia may result in adverse events, including avoidance of procedures; trauma due to fainting or resistance to being restrained; generalized fear of individuals, situations, and objects associated with needles; and avoidance of medical care (McMurtry et al., 2015). Adults who experienced distress during immunizations as children tend to decline dental treatments, avoid regular health care visits, and be nonadherent to preventative measures and medical treatment regimens (McMurtry et al., 2015).

Children receive an estimated 29 immunizations by the age of 6 years (MacLaren & Cohen, 2007). Researchers have identified that approximately 90% of toddlers and 50% of school-age children exhibit severe distress during immunizations and that immunizations are the cause of most health care—associated pain in early life (Taddio et al., 2009). It is essential that health care providers recognize the critical role that uncontrolled pain plays in shaping a child's future experiences with pain and take steps to appropriately manage painful procedures (Taddio et al., 2009).

One approach to pain control during immunizations involves the use of bundle theory, which proposes that using a group of evidence-based interventions together improves patient outcomes. There are numerous bundles used to improve health care delivery, including central line care, Foley catheter care, and ventilator care (Institute for Healthcare Improvement, 2015). Improving health care and patient outcomes through pain prevention during immunizations is equally as important. Bundle theory was used as the framework for this project.

Interventions to Control Pain

There are many interventions that have been found to successfully decrease or eliminate pain during immunizations. These interventions fall into three categories: pharmaceutical, psychological, and physical. Pharmaceutical interventions involve the delivery of medication aimed at controlling pain and can be time consuming and costly. Some of the most commonly used medications for prevention of injection site pain are topical agents, which may not be adequate for the deep tissue pain associated with immunizations given intramuscularly (Shah et al., 2015). Psychological interventions include breathing exercises, distractions, and imagery (Shah et al., 2015). Physical interventions include positioning of the child and the

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specific techniques used to administer the immunization (Shah et al., 2015; Taddio et al., 2009). Although little research has been done to show the effectiveness of psychological and physical interventions used to manage immunization-associated pain, many of the interventions described here have been found to be effective in the management of other sources of procedure-related pain (American Academy of Pediatrics [AAP], Committee on Infectious Diseases, Kimberlin, Brady, Jackson, & Long, 2018).

Psychological interventions

Studies have shown that adequate preparation before a procedure can reduce anxiety and procedural pain for many medical events, including venipuncture (Schechter et al., 2007). This intervention is particularly successful in children older than 2 years. Preparation for immunization should include three components: what will happen (where, how long it will last, and what will be done), how it will feel (pressure, temperature, and level of discomfort to be expected), and strategies to cope with the discomfort (Schechter et al., 2007).

Parental behavior during the immunization has been found to significantly affect the amount of pain and distress the child experiences (Schechter et al., 2007). Empathy, criticism, apologies, giving control to the child, and reassurance have all been found to increase a child's distress during immunizations (Schechter et al., 2007). One study found that 53% of variance in a child's distress was accounted for by maternal behavior. The researchers found that parents trained to provide reassurance were more confident in their ability to help before the procedure than parents trained to provide distraction but that parents who provided reassurance were significantly more upset after the procedure. Children with authoritative mothers (defined as high in warmth and high in control) were less distressed during immunizations than children with mothers who were authoritarian (low in warmth and high in control), permissive (high in warmth and low in control), or unresponsive (low in warmth and low in control; Schechter et al., 2007).

Deep breathing is another tool that can be used to decrease pain during immunizations. Health care providers can help facilitate deep breathing in young children with the use of bubble-blowing or kazoo. These techniques may shift the child's focus away from the injection and provide another form of distraction (Taddio et al., 2010).

One theory that may explain why distraction appears to work well as a pain management strategy is the gate control theory. According to this theory, there is a mechanism in the brain that will act as a gate to increase or decrease impulses that are being received from the peripheral nervous system (Helms & Barone, 2008). This theory suggests that the central nervous system modulates the pain experience; therefore, distraction may affect the processing and perception of pain (Schechter et al., 2007) by using one or more of the five senses to overcome the painful stimuli (Rezai, Goudarzian, Jafari-Koulaee, & Bagheri-Nesami, 2017). Distraction is most effective for children younger than 7 years. Items

that may be used for distraction include movies, party blowers, nonprocedural talk, interactive toys, kaleidoscopes, bubble-blowing, short stories, and music (Schechter et al., 2007). There is evidence that interactive distractions decrease children's pain more effectively than noninteractive distraction (Schechter et al., 2007).

Physical interventions

Several injection techniques have been recommended to reduce injection-associated pain, including recommendations for vaccination site and needle length. Use of the gluteus maximus for pediatric vaccines is no longer supported because of the potential for sciatic nerve injuries. The anterior thigh (vastus lateralis) is the recommended injection site for infants and children through 2 to 3 years old (Advisory Committee on Immunization Practices [ACIP], 2018; AAP, 2018). After 2 to 3 years of age, the deltoid muscle is large enough in most children and is the preferred injection site for children and adolescents (ACIP, 2018; AAP, 2018).

Using the correct needle length is also essential to ensure penetration of the muscle and avoid tissue damage, pain, and other adverse effects (ACIP, 2018; AAP, 2018). The AAP recommends various needle lengths based on age for intramuscular immunizations. For newborns and infants through 12 months of age, a 5/8-inch to 1-inch needle should be used. For toddlers and older children, a 5/8-inch to a 1-inch needle should be used for the deltoid site, and a 1-inch to 1¼-inch needle should be used for injections given in the anterolateral thigh site. After the appropriate site and needle length have been selected, the immunization should be administered by following any special manufacturers' instructions. The needle should be plunged quickly through the skin, followed immediately by injection of the vaccine. Aspiration after needle insertion, which may lead to additional pain, is not necessary because of the absence of large vessels in the recommended injection sites (ACIP, 2018; AAP, 2018).

Simultaneous administration of vaccines refers to the administration of more than one vaccine, at different injection sites, at the same time (AAP, 2018). These injections are often administered by two immunizers. Although there is insufficient evidence to support or refute the use of simultaneous injections versus sequential injections as an effective means to decrease pain (Taddio et al., 2010), there is evidence that the use of simultaneous injections improves immunization rates, and it is recommended by the AAP.

Sucrose water, a sweet solution, has been shown to decrease pain in neonates when administered just before a procedure (Schechter et al., 2007). Sucrose seems to work best during the neonatal period and loses its effectiveness by 4 to 6 months of age. Nonnutritive sucking and breastfeeding have also been shown to provide analgesia when allowed to continue during the procedure (Schechter et al., 2007). Compared with sterile water, sucrose was found to decrease cry time more effectively. Sucrose is believed to work by increasing the serum and cerebrospinal fluid β -endorphin levels when given orally. As the infant's central nervous

system matures, this type of response disappears (Harrison, Beggs, & Stevens, 2012).

Breastfeeding provides a combination of suckling, holding, skin-to-skin contact, and nutritive interventions. Several quasi-experimental studies have shown that breastfeeding before, during, and after immunizations results in a statistically significant decrease in an infant's total cry time (Efe & Ozer, 2007; Kaur, Kaur, Kalra, & Bharti, 2009; Razek & El-Dein, 2009). Additionally, skin-to-skin contact alone has been shown to reduce crying, grimacing, and heart rate during immunizations (Kaur et al., 2009).

Another nonpharmacologic intervention to reduce immunization pain is the use of vibration with cold therapy. This combination helps with the alleviation of sharp, burning, and itching pain (Whelan et al., 2014). Pain control with this approach is achieved through the gate control theory (Whelan et al., 2014). There are three different types of nerves that communicate sensations with our brain: A-delta, A-beta, and C fiber. A-delta fibers are important in detecting pain; they are small and highly myelinated, and they send messages quickly. Their purpose is to get an individual to move away from a painful stimulus. These sensations are easily overcome specifically by a cold stimulus. A-beta fibers are larger and also highly myelinated; these detect motion, which is important to notify individuals if something is touching them, their presence in space, and so on. Finally, C fibers communicate pressure. These fibers are unmyelinated and, therefore, transfer messages more slowly. These messages are constant and will increase in strength if the stimulus progresses or continues. Most importantly, all these fibers meet in the central nervous system and communicate the strongest stimuli to the brain. This is why the motion, cold, and pressure provided by Buzzy can overcome painful stimuli (Baxter, 2014). Researchers have found that the combination of vibration and cold therapy is more effective in decreasing both pain and anxiety associated with venipuncture than other pain interventions (topical agents) or no interventions (Whelan et al., 2014).

Unfortunately, despite recommendations from the AAP (2018), many health care providers do not provide interventions aimed at reducing immunization-associated pain, which may result in short- and long-term adverse effects. The purpose of this project was to implement a sustainable immunization-associated-pain management program for pediatric patients at a primary care clinic located in the rural midwestern region of the United States. The overall goal of this project was to improve the immunization experience for pediatric patients in the primary care clinic. This quality improvement project received institutional review board approval through a large university medical center in the U.S. Midwest. Patient participation in the project was voluntary, and all data were collected without identifying information.

IMPLEMENTATION METHODS

Staff Education

Implementation of the immunization-associated-pain prevention program began with staff education, which was

conducted through one-on-one instruction and demonstration. This education was delivered primarily to the medical assistants who are responsible for administering immunizations. Education included (a) rationale for pain prevention during immunizations, (b) a review of pain prevention strategies, (c) how to help patients and parents to choose age-appropriate pain prevention strategies; (d) essential teaching for parents and older children (including beneficial behaviors for parents/older children and how to help implement pain prevention techniques), (e) appropriate immunization administration techniques, and (f) appropriate techniques for use in a pain-prevention strategy. Hands-on skills training with feedback from the project director was used to teach staff members to use pain prevention techniques. Staff members involved with immunizations were provided with easy-to-use reference sheets for review and use throughout the implementation process. A pain and comfort menu was developed to provide an easy reference for staff to assist families in choosing pain preventions techniques based on age (Figure 1).

Pain Assessment

Before implementation of the pain prevention strategies, baseline data were collected to quantify patients' experience of pain during immunizations, parents' satisfaction with pain management, and staff satisfaction with the use of pain prevention strategies. The Neonatal Infant Pain Scale was used to assess pain in premature and full-term infants and is presented in Figure 2. This previously validated tool is a six-item scale that incorporates one physiologic and five behavioral measures.

Numerous studies have shown that the Faces, Legs, Activity, Cry, Consolability (FLACC) scale is both valid and reliable for use with preverbal children from 2 months to 7 years of age and with cognitively impaired children (Srouji, Ratnapalan, & Schneeweiss, 2010). Each category on the FLACC is rated 0, 1, or 2, with higher scores indicating higher levels of pain. The FLACC scale was used for infants 2 months to 7 years of age and nonverbal cognitively impaired children and is shown in Figure 3. The Neonatal Infant Pain Scale and FLACC scale were completed by the staff member who administered the immunization.

At about the age of 7 years, children can begin to understand the quality of pain. This is when a visual analog or numeric scale can be effectively used. The visual analog scale is a horizontal line that is 100 mm in length with no pain demarcated on the left end and severe pain demarcated on the right (Srouji et al., 2010). This scale was used to measure pain in children older than the age of 7 years. Children were asked to point to the location on the line that corresponded to the pain they were experiencing at three points in time: before, during, and after immunization administration.

Despite self-reported pain being considered the criterion standard of pain assessment, it is not possible to assess self-reported pain in all situations, especially in preverbal, preoperational, and non-English-speaking patients. In instances for which it is not possible to measure self-reported pain, perceived pain is the best available measurement (Zhou, Roberts, & Horgan, 2008). Researchers have found conflicting results in studies examining agreement between self-reported pain scores and perceived pain scores (Zhou, Roberts, & Horgan, 2008) and that

FIGURE 1. Pain & Comfort Menu utilized to help staff and patients/families identify non-pharmacological pain prevention interventions for a particular situation.

Pain & Comfort Menu				
Infants (< 6 months)	6 months – 2 years	2 years – 4 years	4 years – 10 years	10+
Sucrose	Holding	Buzzy w/ ice	Buzzy w/ ice	Buzzy w/ ice
Breastfeeding	Buzzy w/o ice	Deep breathing	Deep Breathing	Deep Breathing
Holding	Distractions	(kazoo, bubbles)	Distraction cards	Distraction
Pacifier	(lights, bubbles, music)	Distraction cards	Music	cards
Buzzy w/o ice	Music	Music	Favorite	Music
Skin-to-skin	Breastfeeding	Positioning	movie/show	Favorite
(kangaroo care)	Comfort item	Comfort item	Positioning	movie/show
Comfort item	(stuffed animals, blanket, etc.)	(stuffed animal, blanket, etc.)	Comfort item	Positioning
(stuffed animal, blanket, etc.)	Book/reading	Book/reading	(stuffed animal, blanket, etc.)	Comfort item
		Pressure	Book/reading	(stuffed animal, blanket, etc.)
			Virtual reality	Book/reading
			Pressure	Virtual reality
				Pressure
				Non-procedural talk

(This figure appears in color online at www.jpmedhc.org.)

FIGURE 2. Neonatal Infant Pain Scale (NIPS). The evaluator picks the appropriate numeric value to quantify the patient’s pain in each category. Scores for each category are added to determine total pain score; the higher the score, the more pain the patient is experiencing.

	Before (1 minute before)	During	After (1 minute after)
Facial Expression 0: Relaxed muscles 1: Grimace			
Cry 0: No cry 1: Whimper 2: Vigorous cry			
Breathing Pattern 0: Relaxed 1: Change in breathing			
Arms 0: Relaxed/restrained 1: Flexed/extended			
Legs 0: Relaxed/restrained 1: Flexed/extended			
State of arousal 0: Sleeping/awake 1: Fussy			
Total			

Adapted from [Lawrence et al. Neonatal Network. 1993;12\(6\):59–66.](#)

parents often underestimate their child’s actual pain. Despite these findings, pain is a source of concern for parents and a potential source of immunization refusal ([MacDonald, Finlay, & Canadian Paediatric Society,](#)

[Infectious Diseases and Immunization Committee, 2013](#)). Therefore, assessing and addressing pain should include parent participation. Parents’ satisfaction was measured using a single question about their satisfaction with the

FIGURE 3. Faces, Legs, Activity, Cry, Consolability (FLACC) scale.

	Before (1 minute)	During	After (1 minute after)
Face 0: no particular expression or smile 1: occasional grimace/frown, withdrawn or disinterested 2: consistent grimace or frown, frequent/constant quivering chin, clenched jaw			
Legs 0: normal position or relaxed 1: uneasy, restless, tense 2: kicking or legs drawn up			
Activity 0: lying quietly, normal position, moves easily 1: squirming, shifting back and forth 2: arched, rigid, or jerking			
Cry 0: no cry/verbalization 1: moans or whimpers, occasional complaint 2: crying steadily, creams or sobs, frequent complaints			
Consolability 0: content and relaxed 1: reassured by occasional touching, hugging or being talked to 2: difficult to console or comfort			

TABLE. Comparing mean pain scores before and after implementation

Mean FLACC pain scores	Preimmunization	During	Postimmunization	Perceived pain score
Patients aged 2 months to 7 years				
Preintervention	0	8.47	3.6	4.73
Postintervention	0	3.77	0.92	3.77
Patients aged 7 years through adolescence				
Preintervention	0	2.9	2.5	2.8
Postintervention	0	1.4	0.7	1.2

immunization experience. Parental response was provided on a 5-point Likert-like scale, where 1 was *completely unsatisfied* and 5 was *completely satisfied*. The same question was used to measure satisfaction in children older than 7 years.

Implementation of Pain Management Strategies

Each immunization session started with age-appropriate parent and child education that included information about the immunization(s) being administered, methods of pain prevention, distraction techniques, and selection of the correct pain prevention strategy. Using the pain and comfort menu, practitioners, parents, and the child worked together to establish an appropriate pain prevention plan. Parents and children were encouraged to use more than one pain prevention intervention for improved pain control. The pain prevention techniques were implemented, and the immunization administered. Data regarding perceived pain, pain scores, and overall satisfaction with the process were collected after the immunization, and parents were provided with educational resources.

Evaluation

Postimplementation pain assessment data were collected using the same tools that were used to collect preimplementation pain assessment data. Descriptive statistics were used to calculate mean pain scores that were compared before and after implementation. The independent-samples *t* test was used to identify statistically significant changes in pain and satisfaction before and after implementation.

RESULTS

During the pre- and postimplementation periods, all patients who required immunizations were older than 2 months. Sample sizes during the data collection period were very small. The preimplementation sample size for patients between the ages of 2 months and 7 years was 15 patients, and the postimplementation sample size for this group was 14 patients. For patients 7 years through adolescence, the preimplementation sample size was 16, and the postimplementation sample size was 7. Immunizations administered for patients 2 months to 7 years of age included influenza, hepatitis A, Pentacel diphtheria, haemophilus influenzae type B, pertussis, polio, and tetanus, PVC13 (pneumococcal pneumonia), DTaP (diphtheria, tetanus, and pertussis), polio, MMR (measles, mumps, and rubella), hepatitis B, Hib (haemophilus influenzae type B), and varicella. Immunizations administered to patients aged

7 years through adolescence included M meningococcal groups A, C, and Y and W-135 polysaccharide diphtheria toxoid conjugate vaccine, influenza, HPV (human papillomavirus), hepatitis A, DTaP, and meningitis B. A comparison of average pain scores before and after implementation is given in the [table](#).

After the implementation of pain prevention techniques, pain scores in both groups were decreased. For children between the ages of 2 months and 7 years, mean pain scores during immunization decreased by 4.7, and mean pain scores after immunization decreased by 2.68. Parents' perceived pain decreased by 0.96. An independent-samples *t* test assuming unequal variances was conducted to compare the pre- and postimplementation scores during and after immunization. An independent-samples *t* test was used to compare the preimplementation pain scores during immunizations with the postimplementation pain scores during immunizations. Preimplementation pain scores during immunizations (mean [*M*] = 8.46) were significantly higher than postimplementation pain scores during immunizations (*M* = 3.76, *t*(24) = 5.24, *p* = .0000224). The preimplementation pain scores after immunizations (*M* = 3.6) were significantly higher than postimplementation pain scores after immunizations (*M* = 0.92; *t* (24) = 4.19; *p* = .000327).

For children 7 years and older, pain during immunization decreased by 1.76, and pain after immunization decreased by 1.94 on average. Parental perceived pain decreased by 1.6. An independent-samples *t* test assuming unequal variance was also conducted on the pain scores for this age group. Preimplementation pain scores during immunizations (*M* = 2.6) were significantly higher than postimplementation pain scores during immunizations (*M* = 1.18, *t*(18) = 2.17, *p* = .043). Preimplementation pain scores after immunizations (*M* = 2.18) were significantly higher than postimplementation pain scores after immunizations (*M* = 0.63, *t*(15) = 2.2, *p* = .043).

Parental satisfaction with pain control did not change between the preintervention group and postintervention group. Parental satisfaction was measured on a Likert-like

After the implementation of pain prevention techniques, pain scores in both groups were decreased.

scale, with 1 being *extremely unsatisfied* and 5 being *extremely satisfied*. Anecdotal evidence showed a universal lack of knowledge of pain prevention interventions and their importance in the preintervention group. This group was not provided with additional education about pain prevention. After the intervention, participants were presented with educational information that included the importance of pain prevention, pain prevention options, and how to implement pain prevention techniques. After the intervention, parents often voiced a lack of awareness of pain control during immunizations before their education.

DISCUSSION

The patients at this clinic clearly experienced a reduction in immunization-associated pain, and anecdotal comments by the nursing staff indicate that they were satisfied with the changes made in immunization practices during the implementation of this project. Unfortunately, despite the strong recommendation for full implementation of new approaches to immunization-associated pain management, the willingness to continue this process, and high satisfaction with this process, this clinic has not continued use of the new pain management approaches.

Immunization-associated pain prevention has not been integrated into the work-flow. Possible explanations for the lack of integration of new practices into the culture could be that the staff responsible for administering immunizations feel a lack of ownership or lack of involvement in the planning process. There is currently no clinic-based champion for this project, and there is a general lack of managerial support: the office manager was not very involved in planning or implementing this project, and the project manager no longer works on the unit. Each of these potential barriers to long-term implementation will need to be addressed to ensure sustainability of this project. Similar projects have found success in integration of nonpharmacologic pain prevention techniques for immunizations into their culture by using superusers or champions with a wide variety of experience, on-the-job training from a peer, and education that not only includes the importance of pain prevention but also the applicable skills (Schurman et al., 2017). Consistent adoption of these techniques appears to be a universal issue despite significant evidence supporting use of various pain prevention techniques. There is limited research surrounding barriers to application. One study reported challenges with application to a diverse population, environmental barriers, and practical issues (inadequate prep time), but in general, more research is needed to illicit more information on the barriers associated with application (Boerner, Gillespie, McLaughlin, Kuttner, & Chambers, 2014).

Methods used to prevent immunization-associated pain are recommended at this clinic and in other settings where immunizations are administered. Despite the lack of sustainability at the implementation site for this project, the children who participated in the pain prevention intervention experienced less pain than those who were immunized without the intervention. Before the implementation of this

project, the use of nonpharmaceutical pain prevention techniques at this clinic was limited primarily to holding and encouraging relaxation of muscles. It is likely that the improvement in pain scores was related to the pain prevention interventions.

Most pain scores were collected by one evaluator, and because of the implementation design, blinding was not possible. This lack of blinding may have resulted in evaluator bias. Efforts were made to decrease this bias using pain scales with established reliability and validity. Data collected to evaluate project effectiveness were also collected from a relatively small sample, which may further limit the reliability of the findings.

Methods used to prevent immunization-associated pain are recommended at this clinic and in other settings where immunizations are administered.

CONCLUSION

Immunization-associated pain often goes unrecognized and is dismissed as short lived. Simple, cost-effective, evidence-based, nonpharmaceutical techniques can decrease the pain associated with immunizations. All health care providers ordering and administering immunizations are encouraged to consider pain control methods. Long-term sustainability could be obtained through effective and frequent communication, involvement of front-line staff, engagement of patients, formalization and standardization of the new process, adequate training, using staff champions, obtaining and sharing measurements of implementation, and celebrating sustainability (Health Quality Ontario, 2013).

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